DSLS Project Pitch

Renk Asik, Leon Igal Inselberg Barbut, Payal Priyadarshini

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1 Project

1.1 Scientific question

Can a multi-omics machine learning model combining gene expression, methylation and phenotype data accurately predict cancer stage?

1.2 Motivation

Cancer prognosis is highly complex, influenced by molecular, epigenetic, and clinical factors. Single-omics approaches often fail to capture this complexity, limiting their predictive power. By integrating gene expression, DNA methylation, and phenotype data, we aim to uncover deeper biological insights and improve prognostic accuracy using machine learning approaches. Such models hold promise for more personalized and effective cancer care.

1.3 Disease/organism/tissue to be studied

This study focuses on breast cancer in humans (Homo sapiens), one of the most prevalent malignancies worldwide. Breast cancer involves complex interactions between genetic, epigenetic, and environmental factors, making it a suitable candidate for multi-omics analysis. The data used in this project include samples derived from breast tissue, incorporating gene expression profiles, DNA methylation patterns, and clinical phenotypes.

1.4 Computational/statistical methods

Statistical analysis will be performed using R and python software.

1.4.1 Preprocessing

For **RNASeq**: Remove sex chromosomes, exclude genes with low variation to reduce dimensionality and retain biologically relevant variables. Map EnsembleID to genes for biological interpretability.

For **methylation data**: Remove probes with missing data and probes that match the sex chromosome. Remove overlapping features, keep SNPs with 5% allel frequency. Filter out low expression levels using count per million (CPM) method.

Missing values for all types of data will be imputed using appropriate statistical methods.

1.4.2 Stratification

To assess whether our model performs consistently across subgroups, we will stratify patients by clinical variables such as:

- Age group
- Ethnicity

1.4.3 Statistical analysis

Unsupervised learning methods, such as **principal component analysis (PCA)**, **t-SNE** and **UMAP** will be applied for the exploratory analysis to visualize sample structure and detect potential clusters or batch effects across the multi-omics data. These methods reduce high-dimensional omics data into interpretable 2D/3D representations.

DeSeq2 will be used to detect **differentially expressed genes (DEGs)** from RNA-Seq data.

LIMMA, *getMethylDiff* and *calculateDiffMeth* will be used to find **differential methylated sites** from methylation data. We will use a linear modeling approach *lmFit* followed by empirical Bayes moderation (*eBayes*). Significant features will be extracted using *topTable* and multiple testing correction will be applied using FDR-adjusted p-values.

1.4.4 Machine learning

For model development, we will employ supervised learning algorithms including but not limited to **random forests**, **support vector machines (SVMs)**, and **logistic regression**. Model performance will be evaluated using cross-validation (e.g. 10-fold) and metrics such as accuracy, AUC-ROC, and F1-score. We will use cancer stages of patients as labels.

1.5 Annotation of Results

We will annotate the differentially expressed genes (DEGs) and differentially methylated sites (DMSs) using publicly available biological databases such as:

- Gene Ontology (GO): to identify associated biological processes, molecular function, cellular components.

- KEGG Pathways: to contextualize genes within biological pathways
- biomaRt (R): for mapping gene IDs and obtaining functional annotations

This annotation will help us interpret the biological relevance of our findings and identify key mechanisms involved in breast cancer prognosis.

1.6 Related work

Yang, M. et al. [3] aims to improve colorectal cancer (CRC) survival prediction by identifying biomarkers across mRNA, miRNA, and tissue microbiome data. Unlike prior single-omics approaches, it uses a multi-omics strategy to enhance personalized therapy for CRC patients.

Mohammed, TIA et al. [1] presents a deep learning model for lung adenocarcinoma (LUAD) detection by integrating mRNA, miRNA, and DNA methylation data. Using PCA for feature reduction and SMOTE for class balancing, the combined omics data were classified with a CNN, achieving high predictive performance with 97% accuracy, precision, recall, and F1-score.

Poirion, O.B. et al. [2] introduces DeepProg, an ensemble framework combining deep learning and machine learning to integrate multi-omics data for robust cancer survival subtype prediction. It outperforms existing methods in risk stratification across multiple cancers and links poor survival subtypes to shared genomic signatures related to the extracellular matrix, immune dysfunction, and mitosis.

1.6.1 Difference from related work

While prior studies have applied multi-omics integration to cancers like colorectal (Yang et al.), lung adenocarcinoma (Mohammed et al.), and pan-cancer settings (Poirion et al.), our project focuses specifically on breast cancer prognosis using gene expression, DNA methylation, and clinical phenotype data from the same patient cohort.

Unlike mentioned research papers, who used deep learning, we apply classical supervised models (e.g., random forests, SVMs, logistic regression).

2 Data

2.1 Where will you get your data from?

We will get the data from The Cancer Genome Atlas (TCGA), a comprehensive and publicly available resource that provides multi-omics data across various cancer types. The data will be accessed through the UCSC Xena platform.

2.2 Samples

The study will include around 1200 breast cancer patient samples from TCGA. We will focus on samples that have complete data across gene expression, DNA methylation, and phenotype to ensure robust multi-omics integration. Sample selection criteria will include availability of survival outcomes or cancer grade and relevant clinical annotations to enable accurate prognosis prediction.

2.3 Is the data multi-modal (multi-omics)? Are all data coming from the same study?

Yes, the data used in this study is multi-modal, integrating gene expression, DNA methylation, and clinical phenotype information to capture different biological layers influencing cancer prognosis. The multi-omics datasets and clinical annotations originate from the same cohort of breast cancer patients. This consistency allows for reliable integration and analysis across data types.

2.4 Format availability

A fully preprocessed version of the data is not currently available. Significant preprocessing will be required before analysis, including filtering out sex chromosome genes/probes, removing low-expression or low-variation features, and handling missing values through statistical imputation. However, the data is already normalized.

2.5 Data access restrictions

The data from The Cancer Genome Atlas (TCGA) are publicly available.

3 Summary of project plan

This project aims to develop a multi-omics machine learning model to predict cancer stage in breast cancer patients by integrating gene expression, DNA methylation, and clinical phenotype data. Breast cancer is a highly prevalent and biologically complex disease, making it an ideal case for multi-modal analysis. By combining molecular and clinical features, we seek to improve prognostic accuracy and contribute toward more personalized cancer care.

We will source our data from The Cancer Genome Atlas (TCGA) via the UCSC Xena platform, focusing on breast cancer patient samples with complete data across RNA sequencing, methylation, and phenotype modalities. The datasets originate from the same cohort, ensuring consistency and reliability in integration.

Preprocessing will be required, as a fully preprocessed dataset is not available. Steps will include filtering out sex chromosome genes/probes, removing features with low expression or variation, handling missing values via imputation.

Exploratory analysis will involve unsupervised learning techniques such as PCA, t-SNE or UMAP.

For prediction, we will employ supervised machine learning models including random forests, support vector machines (SVMs), and logistic regression. These models will be trained on integrated multi-omics features and evaluated using cross-validation and standard metrics, using cancer stage as the outcome label.

4 Timeline

Week 1: 9–15 June: Finalize data acquisition and integration (preprocessing, quality control, data transformation).

Week 2: 16–22 June: Conduct statistical analyses (differential expression, methylation analysis, functional enrichment). Begin machine learning model development.

Week 3: 23–29 June: Complete machine learning pipeline and finalize visualizations.

Week 4: 30 June–6 July: Begin biological annotation of results and start preparing presentation slides.

Week 5: 7–13 July: Finalize annotation and complete presentation materials.

Week 6: 14–20 July: Deliver presentation (early in the week) and submit written report draft (17–20 July).

Week 7: 21–27 July: Finalize and polish the written report for submission.

References

[1] Tehnan IA Mohamed and Absalom El-Shamir Ezugwu. Enhancing lung cancer classification and prediction with deep learning and multi-omics data. *IEEE Access*, 12:59880–59892, 2024.

[2] Olivier B Poirion, Zheng Jing, Kumardeep Chaudhary, Sijia Huang, and Lana X Garmire. Deepprog: an ensemble of deep-learning and machine-learning models for prognosis prediction using multi-omics data. *Genome medicine*, 13:1–15, 2021.

[3] Min Yang, Huandong Yang, Lei Ji, Xuan Hu, Geng Tian, Bing Wang, and Jialiang Yang. A multi-omics machine learning framework in predicting the survival of colorectal cancer patients. *Computers in biology and medicine*, 146:105516, 2022.